

Training Day 10 Report

Date: 26 June 2025

Module Name: Network Scanning with Nmap & Packet Analysis with Wireshark

Mode: Practical

Main Focus of the Module

Day 11 involved the installation and in-depth hands-on use of **Nmap** and **Wireshark**, two of the most powerful tools in network scanning and packet analysis. These tools were used to explore the structure and security posture of target domains. We focused on **active scanning**, **port discovery**, **OS fingerprinting**, and **network traffic monitoring**.

Topics Covered

Installation of Nmap

Nmap (Network Mapper) is a free, open-source tool for:

- Host discovery
- Port scanning
- Service enumeration
- OS detection

Installation on Kali Linux:

```
bash
Copy code
sudo apt update
sudo apt install nmap
```

Installation on Windows:

- Download from: <https://nmap.org/download.html>
-

Common Nmap Commands Practiced

Command	Function
<code>nmap [domain/IP]</code>	Basic scan
<code>nmap -sS [target]</code>	Stealth SYN scan

Command	Function
<code>nmap -sV [target]</code>	Service version detection
<code>nmap -O [target]</code>	OS fingerprinting
<code>nmap -A [target]</code>	Aggressive scan (OS, services, traceroute, scripts)
<code>nmap -F [target]</code>	Fast scan of top 100 ports

Examples Practiced:

```
bash
Copy code
nmap facebook.com
nmap -A testphp.vulweb.com
```

Findings:

- Open ports, potential server software used, and basic fingerprinting of web-facing servers

🎯 Target Domains Used for Practice

- `facebook.com` – Real-world scanning with restricted/safe options
- `testphp.vulweb.com` – Vulnerable test site used for aggressive scanning
- `scanme.nmap.org` – Official Nmap practice target

🔧 Installation of Wireshark

Wireshark is a GUI-based network protocol analyzer that captures and displays packets in real-time.

Installation on Kali Linux:

```
bash
Copy code
sudo apt install wireshark
```

Installation on Windows:

- Download from: <https://www.wireshark.org/download.html>

🔧 Wireshark Hands-On Activities

Steps:

1. Selected `eth0` interface to capture live traffic
2. Visited domains: `facebook.com`, `vulweb.com`
3. Captured HTTP, DNS, and ICMP packets
4. Applied filters:
 - o `http` → To see web requests
 - o `ip.addr == 192.168.1.1` → Filter by IP
 - o `tcp.port == 80` → Filter by port

Analysis Performed:

- Observed TCP three-way handshake
- Viewed packet headers: source/destination IPs, flags, payloads
- Interpreted DNS lookups and HTTP GET/POST requests

Key Takeaways

- **Nmap** helps identify exposed services and network architecture quickly
- Learned differences between **passive** (Wireshark) and **active** (Nmap) techniques
- Gained experience using filters to isolate meaningful packets in Wireshark
- Understood that unauthorized aggressive scanning on real domains is unethical/illegal without permission
- Practicing on **safe targets** (like `scanme.nmap.org`) is essential for skill-building